

Ethernet and Fast Ethernet

Module 23

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0:03 [Music]

0:10 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:18 University of Science and Technology Hisar harana I welcome you all to the

0:25 video lecture series of cyber security and information secur in this lecture we will study about

0:33 ethernet and fast ethernet which is basically used for data transmission on

0:40 wired Lan at a speed of 10 and 100 Mbps respectively first of all please see the

0:49 contents which I shall cover after covering brief introduction about ethernet I shall take up protocol

0:58 architecture and frame format of ethernet which will be followed by Mac operations in the

1:07 ethernet then I shall take up fast ethernet and protocol architecture of fast

1:14 ethernet so now let us start ethernet is a family of computer networking

1:21 Technologies for local area network and metropolitan area network it was

1:28 developed as an experimental coxl cable network in 1970s by jerox

1:37 Corporation initially it operated with a data rate of just 3 Mbps using a carrier

1:44 sense multiple assess with Collision detection protocol for Lan it was

1:51 commercially introduced in 1980 and standardized in 1983 as i e 8

2:01 2 minutes, 1 second 2.3 it has since been refined to support higher bit rates and longer link

2:09 distances over time ethernet has largely replaced competing wired land

2:16 Technologies such as token ring token bus fddi and

2:23 arknet ethernet land topology is currently the most common Network architecture

2:30 ethernet topologies are generally bus or bus star topologies the I 802.3 standard

2:40 defines ethernet protocols for OSI media access control sublayer and physical

2:48 layer Network characteristics the name ethernet refers to the cable ether four types of

2:57 cabling server ethernet as shown in table and figure
3:04 table shows four types of cable standards however figure part A shows 10
3:12 base five connections Part B shows 10 base 2 connections and part C shows 10
3:20 base T connections historically 10 base 5 cabling popularly called thick ethernet
3:29 came first connections to it are generally made using vampire tabs as shown in part
3:37 A of figure in which a pin is very carefully forced halfway into the coaxial
3:45 core in Ethernet standard the first number is the speed in
3:53 Mbps then comes the word base to indicate base band transmission
4:01 4 minutes, 1 second accordingly the notation 10 base 5 means that it operates at 10
Mbps using
4:10 baseband signals and can support segments of up to 500
4:17 M the second cable type 10 base 2 was made up of thin coaxial cable using BNC
4:26 connectors to form P Junctions rather than using vampire tabs as shown
4:34 in Part B of figure thin ethernet is much cheaper and
4:41 easier to install but it can run for only 185 m per segment Each of which can
4:50 handle only 30 machines problem with 10 base 5 and 10
4:57 Bas 2 is that whole network runs on single Cable in case cable gets fault at
5:06 some place whole network goes down the problems associated with
5:14 finding cable brakes drove systems toward a different kind of wiring
5:20 pattern in which all stations have a cable running to a central Hub this
5:28 scheme is called 10 Bas T in this scheme each computer is connected to HUB
5:36 through separate twisted pair cable as shown in part C of figure it supports
5:44 100 m distance with 1024 nodes and is cheapest system a
5:52 fourth cabling option for ethernet is 10 Bas F which uses fiber Optics this
6:01 6 minutes, 1 second alternative is expensive due to the cost of the connectors and
Terminators but it has excellent noise
6:11 immunity and is the method of choice when running between buildings or widely
6:18 separated hubs runs of up to 2,000 M are allowed
6:25 it also offers good security since wire tapping a fiber is much more difficult
6:33 than wire tapping a copper wire it supports 1024 nodes and is best
6:41 suitable for connecting two buildings now let us see ethernet
6:49 layered protocol architecture in Ethernet both the data link and the
6:55 physical layers are involved in the creation and transmission of frames the
7:03 physical layer is related to the type of Lan cabling and how the bits are
7:09 transmitted and received on the cable the data link layer is divided into Su
7:16 layers which are logical link control LLC and media access control Mac the LLC
7:26 sublayer is responsible for identifying and passing data to the network layer

7:33 protocol and MAC sub layer is responsible for medium access control
7:39 for shared data the MAC sublayer address is the physical Hardware address of the
7:46 source and destination computer this sublayer controls which
7:53 computer devices send and receive the data and allows them to communicate
8:00 with the physical layer the frame format of IEEE
8:08 802.3 is as shown in figure the Preamble of the frame which is of 7 bytes
8:16 indicates the start of a new frame and establishes synchronization conditions
between
8:25 devices the last byte of Preamble known as start of frame delimiter (SFD) always has
8:33 a 10101010 bit pattern this byte indicates
8:42 the start of a frame the destination address of 6 bytes is the hardware MAC
8:49 address of the receiving device and the source address of 6 bytes specifies the
8:57 hardware MAC address of the sending device the length field of 2 bytes
9:05 indicates the length of frame the data field is for the actual data being
9:12 transmitted from device to device and length of data field is 46 to 1500
9:21 bytes in case length of data is less than 46 bytes padding is done to make it
9:28 46 bytes the cyclic redundancy check (CRC) of four bytes checks that the frame is
9:38 received free from error after frame format let us see how
9:45 medium access control works in Ethernet protocols in which stations
9:53 listen for a carrier and act accordingly are called carrier sense protocols
10:00 Ethernet uses CSMA/CD as medium access control protocol for channel
10:08 sensing when a station has data to send it first listens to the channel to see
10:15 if anyone else is transmitting at that moment if the channel is busy the
10:23 station waits until it becomes idle when the station detects an idle channel it
10:31 transmits a frame if a collision occurs the station waits a random amount of
10:37 time and starts all over again the process of CSMA/CD is shown
10:46 in flowchart in this flowchart when a station is ready to send data it senses
10:53 the channel if channel is busy it makes a new attempt to sense the channel if
11:01 11 minutes, 1 second channel is free it transmits data and senses the channel for
Collision in case collisions are
11:10 detected it transmits a jam signal to indicate that Collision has occurred and
11:17 it waits as per backoff strategy if no collisions are detected
11:26 transmission completes CSMA/CD uses the conceptual model
11:34 shown in figure CSMA/CD can be in one of three states contention transmission
11:44 or idle at the point marked t_0 a station has finished transmitting its
11:53 frame any other station having a frame to send may now attempt to do so if two
12:01 12 minutes, 1 second or more stations decide to transmit simultaneously there will
be a

12:08 collision collisions can be detected by looking at the power or pulse width of
12:15 the received signal and comparing it to the transmitted signal after a station
12:23 detects a collision it aborts its transmission waits a r period of time
12:30 and then tries again assuming that no other station has started transmitting
12:38 in the meantime therefore our model for CSMA by CD will consist of alternating
12:47 contention and transmission periods with ideal periods occurring when all
12:54 stations are quiet now let us see what happens if Collision occurs after
13:03 the first Collision each station waits either zero or one slot times before
13:10 trying again if two stations collide and each one picks the same random number
they
13:19 will collide again after the second Collision each one picks either 0 or 1 2
13:28 or three at random and waits that number of slot
13:35 times if a third Collision occurs then again the number of slots to wait is
13:42 chosen at random from the interval 0 to
13:47 $2^3 - 1$ that is 7 in journal after 1
13:54 collisions a random number between zero and $2^k - 1$ is chosen and that
14:04 number of slots is skipped however after 10 collisions have
14:10 been reached the randomization interval is frozen at maximum of
14:16 1023 slots after 16 collisions the controller throws in the towel and
14:24 reports failure back to the computer further recovery is up to the higher
14:32 layers this algorithm is called binary exponential back off now let us see the
14:41 reason for having padding bits in Ethernet frame reason for having a minimum
length
14:49 frame is to prevent a station from completing the transmission of a short
14:56 frame before the first bit has even reached the far end of the cable where
15:03 it may collide with another frame this problem is illustrated in
15:09 figure in figure a at Time Zero station A at one end of the network sends off a
15:18 frame let T be the propagation time for this frame to reach the other end B in
15:27 figure B just before the frame gets to the other end that is at time T minus ϵ
15:35 the most distant station B starts transmitting as shown in figure C when B
15:43 detects that it is receiving more power than it is putting out it knows that a
15:50 collision has occurred so it aborts its transmission
15:56 and generates a 40 8bit noise burst to warn all other stations in other words
16:06 it jams The Ether to make sure the sender does not miss the collision at about time
16:16 $2T$ the sender sees the noise burst and aborts its transmission as shown in
16:23 figure D it then waits a random time before trying
16:29 again if a station tries to transmit a very short frame and collision occurs

16:36 the sender will then incorrectly conclude that the frame was successfully
16:44 sent to prevent this situation from occurring all frames must take more than
16:50 2T time to send data frame so that the transmission is still taking place when
16:58 and the noise burst gets back to the sender that is the reason why padding
17:06 bits are added so that transmission is not complete in 2T time frames with
17:14 fewer than 64 bytes are padded out to 64 bytes with a pad
17:22 field so this was about ethernet now let us move on to fast ethernet fast
17:30 ethernet is traditional csma by CD at 100 Mbps over twisted pair cable the
17:40 idea of fast ethernet was first proposed in 1992 during the early development of
17:49 fast ethernet two different groups worked out standard proposals and both
17:55 were finally approved but under different title committees one standard
18:03 passed a full review in June 1995 and was formally assigned the name
18:12 802.3 U and the other became 100 VG any
18:18 Lan which is now governed by IEEE 802.2
18:24 committee the latter uses the demand priority media access method instead of
18:31 CSMA by CD method 100 VG any Lan has not caught on
18:40 while fast ethernet enjoyed great success in Enterprise Lan
18:48 environment due to slow speed of ethernet fast ethernet came into existence which
is simple speed up
18:56 version of ethernet with same rules and regulations the 802.3 committee decided
19:06 to go with a speed up ethernet for three primary reasons one the need to be
19:14 backward compatible with existing ethernet Lan two the fear that a new protocol
19:23 might have unforeseen problems three the desire to get the job done before the
19:32 technology changed in fast ethernet instead of creating a completely new
19:39 protocol the committee decided to keep all the old packet formats
19:46 interfaces and procedural rules and simply reduce the bit time
19:53 from 100 nsec to 10 nsec this effectively increased the
20:00 bandwidth to 1 Gbps the protocol was officially known as
20:08 802.3 U but was more commonly called Fast ethernet the change in bit time
20:17 presented a number of challenges to the designers of fast ethernet the reduction
20:24 in bit time increases the number of bits sent within a set time period however these
20:33 bits still take the same amount of time to travel across a length of wire that
20:40 is the propagation delay is the same over a given length this means that more
20:48 bits will be sent during the propagation delay time used as the Collision
20:56 window either minimum frame size need to be increased or the propagation
delay need to be reduced by reducing
21:06 cable length changing the minimum frame size would have caused problems with a

backward compatibility of this standard so the maximum Network size was reduced with typical maximum cable length of 100 m now let us see different media types supported by fast ethernet fast ethernet defines three basic physical layer specifications as shown in figure 100 base T4 this uses Category 3 four pair twisted pair cable it supports a distance of 100 m and RJ 45 connectors 100 Bas TX this uses Category 5 twisted pair which are more expensive than Category 3 cables these are Cat 5 UTP or type 1 STP cable with RJ45 connectors this scheme allows full duplex communication stations can trans transmit at 100 Mbps and receive at 100 MVPs at the same time 100 base FX this uses two strands of 62.5 or 125 multimode fiber one for each Direction so it to is full duplex with 100 Mbps in each Direction in addition the distance between a station and a hub can be up to 2,000 M it supports SC mic or STD connectors there are also few other physical standards in fast ethernet which will be explained now 100 base SX it is a version of fast ethernet over or optical fiber it uses two strands of multimode optical fiber for receive and transmit it is a lower cost alternative to using 100 base effects because it uses short wavelength Optics which are significantly less expensive than the long wavelength Optics used in 100 base FX 100 base SX can operate at distances up to 550 M 100 base BX it is a version of fast ethernet over a single strand of optical fiber single mode fiber is used along with a special multiplexer which splits the signal into transmit and re receive wavelengths the two wavelengths used for transmit and receive are 131 nanometer and 1 1550 nanometer distances supported are 10 20 or 40 kilom 100 base LX 10 it is a version of fast ethernet over two single mode optic fibers it has a nominal reach of 10 kilm and a nominal wavelength of 1310 nanometer now let us move on to layered protocol architecture of fast ethernet various layers of the fast ethernet protocol architecture are shown in figure the Mii is medium independent interface between the Mac layer and the physical layer it allows any physical layer to be used with the Mac layer the Mii provides two media status signals one indicates presence of the carrier and the other indicates absence of collision the reconciliation Su layer

25:54 Maps these signals to physical signaling primitiv understood by the existing map
26:03 the physical layer is divided into three sublayers physical coding sublayer PCS
26:11 this sublayer provides a uniform interface to the reconciliation layer for
26:17 all physical media carrier sense and collision detect indications are
26:24 generated by this sublayer it also manages the autonegotiation
26:31 process by which the network interface card communicates with the network to
26:38 determine the network speed 10 or 100 MVPs and mode of operation whether it is
26:47 half duplex or full duplex physical medium attachment PMA
26:54 this sublayer provides a medium indep dependent means for the PCS to support
27:02 various serial bit oriented physical media this layer serializes code groups
27:10 for transmission and deserializes bits received from The
27:16 Medium into code groups physical medium dependent PMD this Maps the physical
27:26 medium to the PM it defines the physical layer signaling used for various
27:35 media so this was about layered protocol architecture of fast
27:41 ethernet frame format of fast ethernet is same as that of
27:48 ethernet so friends this was ethernet and fast ethernet finally summarizing we
27:56 learned about introduction to ethernet which was followed by layered protocol
28:03 architecture and details of physical and Mac operations in Ethernet after
28:10 ethernet we discussed fast ethernet hope the concepts explained in
28:18 this lecture were understandable and helpful hope to see you again in the
28:25 next lecture till then goodbye bye enjoy the day thank you
28:32 [Music]